



GEN AI KICKSTARTER · PROJECT 01

INSTRUCTIONS

Personal Research Agent

An always-ready research agent that answers across your PDFs, links, and videos — with a source behind every line.

Day 5 · Perplexity · NotebookLM · Claude

Your turning point in the AI era.

01

Personal Research Agent

The brief — what you're building and why

WHAT YOU'LL BUILD

A single assistant that has read everything you give it — PDFs, article links, and YouTube videos — and answers any question about that material in seconds, always pointing to the exact source. Instead of re-reading ten tabs, you ask; it answers, with citations.

- ✓ Answers grounded in your own sources, not the open internet
- ✓ Pulls YouTube transcripts automatically — no manual note-taking
- ✓ Cites the exact passage behind every claim
- ✓ Tops up with live web data for anything newer than your sources

WHY IT MATTERS

Research is where most beginners lose hours. This build turns a scattered pile of material into one trustworthy assistant — the single highest-leverage study-and-work habit. It also teaches grounding and citations: the skill that separates useful AI from confident guessing.

WHERE IT RUNS

Three free tools working together: NotebookLM (your private, source-grounded notebook), Perplexity (live, cited web search), and Claude (the synthesiser that merges them).

WHEN TO BUILD & USE IT

Day 5 of the Kickstarter. Build it once; use it every time a topic has more than a couple of sources — before an exam, a meeting, a purchase, or a piece of writing.

HOW TO APPROACH IT

The method — *step by step*

These are the moves, not the answers. Work through them in order; the worked solution lives in the separate Solution guide.

- 1 Choose ONE topic you actually care about and gather 8–10 sources across three types: PDFs, article links, and YouTube videos.
- 2 Set up a source-grounded notebook and load every source into it — notice how it handles each type differently.
- 3 Learn to ask questions that force citations, so you can trust and trace every answer.
- 4 Separately, run your time-sensitive questions through a live web tool and keep only what's cited.
- 5 Bring both sets of answers into Claude and design a prompt that merges them into one clean, cited brief.

REQUIREMENTS

- ✓ One focused topic
- ✓ 8–10 sources incl. at least one YouTube video
- ✓ Citations visible on every answer
- ✓ A one-page synthesised brief as the output

HAND IN

Your notebook + a one-page, fully-cited research brief on your topic.

A LITTLE HINT

You don't need the perfect question first try. Ask a broad one, read the cited answer, then ask a sharper follow-up. Three rounds beats one long prompt.